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MAcWorP – Massively Accessible Workflow Platform

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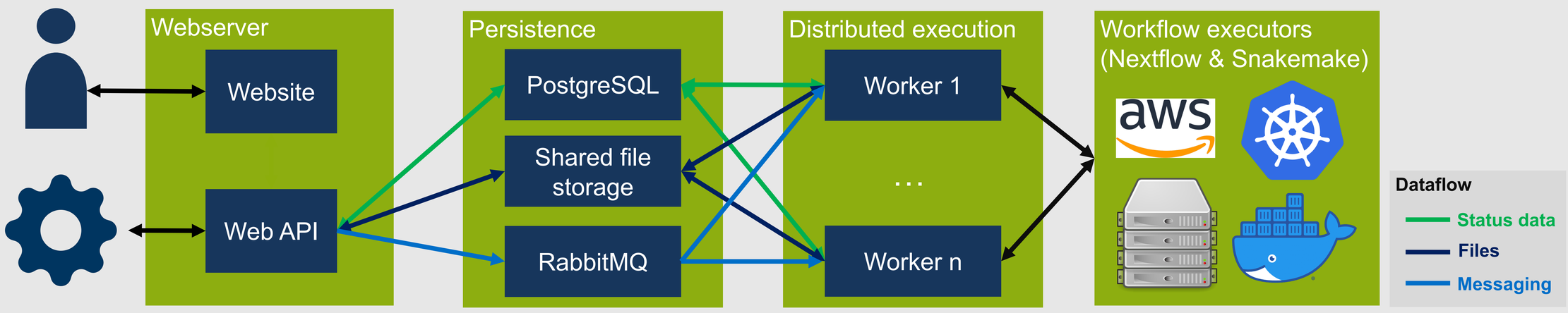
Abstract

When analyzing data in life sciences it is often necessary to use a variety of tools to get the most insight and comprehensive results. To achieve this in an easy and reproducible way, workflow engines like Nextflow or Snakemake are used to build sophisticated data analysis pipelines. While those engines are powerful tools for developers, they are a high obstacle to non-technical users due to their unfamiliarity to the command line or to Unix-like operating systems.

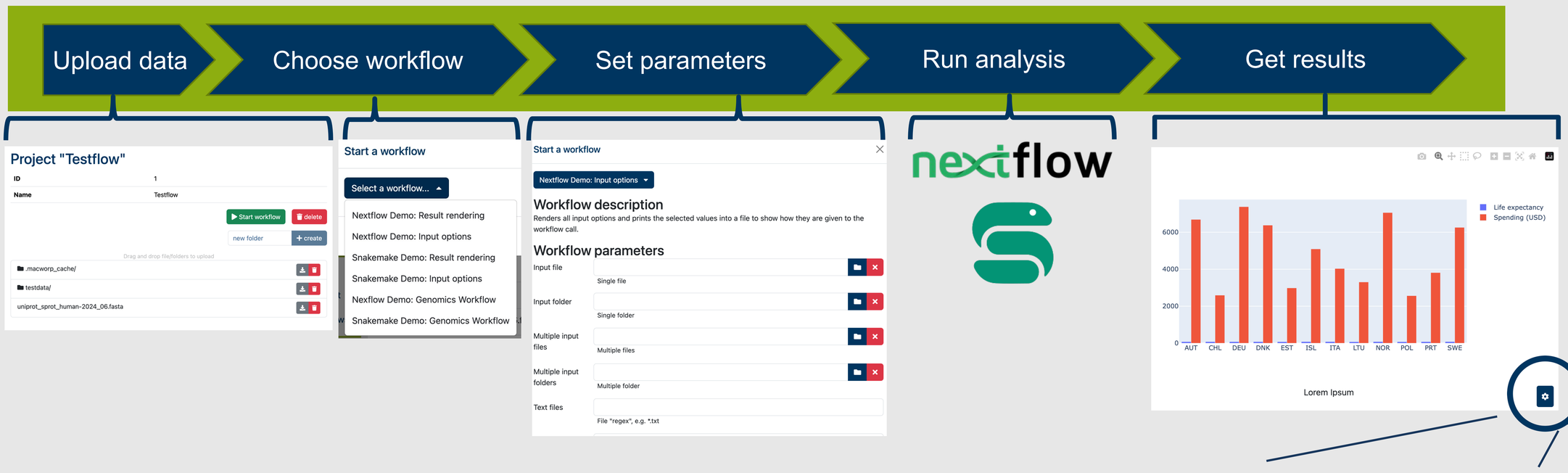
The Massively Accessible Workflow Platform (MAcWorP) is designed to overcome these limitations by making powerful data analysis pipelines accessible also to less technically affine users via a web interface independent of the underlying workflow engine. Users can upload their data and run the embedded workflows on the platform using a simple web browser. Furthermore, any result of the workflows can be visualized directly inside the browser as images, tables, interactive plots and more.

MAcWorP enables developers to quickly deploy Nextflow and Snakemake workflows on a web-based platform which creates a graphical user interface for the workflow parameterization, execution and visualization of results. MAcWorP is open source and freely available at https://github.com/cubimedrub/macworp.

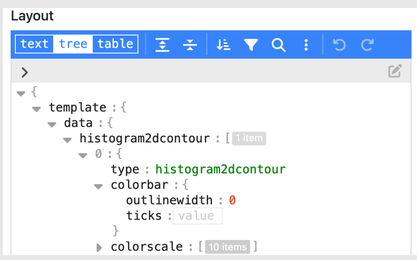
Architecture



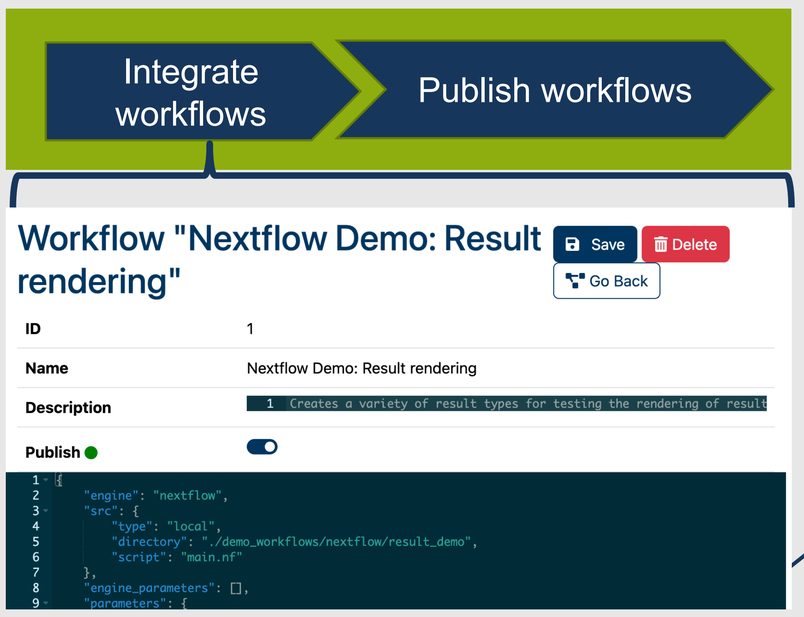
For users



Interactive plots can be adjusted before downloading, e.g. to match journal rules.



For developers



Workflows are defined by a validated JSON object, which defines the user input elements, the structure of the result view and parameters of the workflow engines.

Upcoming

- User access management
- Per project
  - Store results for multiple sequential workflow runs
  - Archive the settings for each workflow run

Conclusion

- Graphical user interface for workflows
- OS independent
- Customizable/downloadable/interactive result plots
- Supports Nextflow and Snakemake
- Input validation
- Web-API to include or automate workflow execution
- Self hostable / freely available as open-source
- FAIR

Find MaCWorP @ https://github.com/cubimedrub/macworp

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MAcWorP was designed and partially developed for an ELIXIR implementation study of the Proteomics Community